

SOURAV ROY

Quantitative Research | Risk Analytics | Data Science

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GitHub: github.com/Souravfrp | Portfolio: souravfrp.github.io | arXiv: 2508.19582

PROFILE

Master's graduate in Mathematics and in Computer and System Sciences from the Tata Institute of Fundamental Research (TIFR). My research background is in convex geometry, probability, randomized algorithms and optimization, with emphasis on proofs, error bounds and complexity. I apply the same evidence-led approach in Python to financial time series, risk forecasting, portfolio evaluation and fraud-risk data analysis. I am seeking quantitative research, risk analytics and data science roles.

EDUCATION

M.Sc., Computer and System Sciences | 2021-2026

Tata Institute of Fundamental Research, School of Technology and Computer Science, Mumbai

M.Sc., Mathematics | 2019-2021

Tata Institute of Fundamental Research, Centre for Applicable Mathematics, Bengaluru

B.Sc. (Honours), Mathematics | 2015-2018

Serampore College, University of Calcutta

RESEARCH PAPER / PREPRINT

Approximating Mixed Volumes to Arbitrary Accuracy | 2025

Hariharan Narayanan and Sourav Roy | arXiv:2508.19582 | revised December 2025

- Co-authored a randomized approximation algorithm for mixed volumes of convex polytopes given by integer vertices, combining convex optimization, Lorentzian polynomials and polytope subdivision.
- Established multiplicative approximation and failure-probability guarantees, with polynomial runtime for a fixed number of distinct input polytopes under the stated assumptions.

MASTER'S THESIS

Approximation of Mixed Volumes of Special Convex Bodies | March 2026

TIFR, Mumbai | Supervisor: Prof. Hariharan Narayanan

- Studied mixed volumes of a polytope and a Euclidean ball through the Steiner polynomial, geometric decomposition, sampling and log-concavity.
- Developed and analysed noisy binary search for a scaling parameter, including decision-error and runtime bounds; separated the idealized oracle model from practical approximate optimization.

PROJECTS

Quant AI Market Lab | Independent quantitative research | Completed v1.0.0 | 2026

Python, NumPy, pandas, SciPy, scikit-learn, CVXPY | github.com/Souravfrp/quant-ai-market-lab

- Built a validated eight-ETF workflow covering returns, covariance and PCA; verified return identities, matrix properties and eigendecomposition. The first three PCs explained 79.01% of standardized historical return variation.
- Compared KMeans and Gaussian HMM market-condition models, then evaluated Ridge and Random Forest forecasts of next-five-day SPY return RMS on 565 common chronological dates (Ridge MAE 0.003201 versus 0.003443).
- Evaluated long-only minimum-variance portfolios across five covariance methods, three transaction-cost assumptions, nine benchmark scenarios and 32 strategy-period robustness checks; documented timing and retrospective limitations.

Quant Market Lab II - Prospective Risk Forecasting | 2026 | In progress

Python, pandas, NumPy, scikit-learn, unittest | github.com/Souravfrp/quant-market-lab-ii

- Designed a fixed-cutoff SPY risk-forecasting experiment using only information through 31 August 2026; implemented Ridge, constant and rolling-RMS baselines across four lead times and preserved 12 versioned September-December forecasts.
- Stored model, data and source fingerprints with each release and added target-timing and prefix-invariance tests so post-cutoff observations cannot alter frozen forecasts. September is labelled retrospective; later windows are evaluated only after realization.

PaymentGuard | Payment-fraud risk study | 2026 | In progress

Python | github.com/Souravfrp/payment-guard

- Checksum-verified and validated 1.09 million AMLNet v2 transactions in 50,000-row chunks; audited 17 top-level columns and 47 nested metadata paths in a 729 MiB synthetic dataset.
- Detected 4,527 chronological decreases in file order and target-revealing metadata covering all 1,411 fraud labels; produced a documented feature policy for chronological modelling. Cleaning, modelling, SQL and the approve-review-block decision policy remain in progress.

ResearchAudit | Multi-agent validation proposal | 2026

Python | github.com/Souravfrp/research-audit

- Defined an evidence-based audit plan and froze Quant AI Market Lab v1.0.0 as the first versioned target; multi-agent implementation and audit experiments have not yet begun.

RSA Encryption and Decryption in Python | 2019; revised September 2026

INSPIRE Summer Project, Presidency University, Kolkata | Guide: Dr. Abhishek Adhikari | github.com/Souravfrp/rsa-educational

- Implemented educational RSA key generation and ASCII encryption/decryption using Miller-Rabin testing, the extended Euclidean algorithm and modular exponentiation.
- Added UTF-8 support, input validation and seven unit tests in the 2026 revision; documented the correctness argument and a public-key lookup attack demonstrating the limits of byte-wise textbook RSA.

ADDITIONAL RESEARCH PROJECT

Exploring the Efficient Computation of Permanents of Graph Laplacians | December 2022

TIFR | Supervisor: Prof. Piyush Srivastava | Public project report

- Proved the permanent analogue of Brenner's result for diagonally dominant matrices (Theorem 3.5 of the report).
- Explored tensor-product formulations and recursive expressions for paths, cycles and trees, and surveyed Pfaffian methods connecting selected permanent computations to determinants.

TECHNICAL SKILLS

Programming and scientific computing: Python, NumPy, pandas, SciPy, scikit-learn, CVXPY; C, MATLAB, SageMath, Mathematica.

Methods: Probability, convex optimization, randomized algorithms, sampling, complexity analysis, PCA, regression, clustering, hidden Markov models, chronological validation, fixed-cutoff forecasting, portfolio backtesting, leakage auditing and numerical validation.

Tools: Git, GitHub, Jupyter, LaTeX, Python unittest.

SELECTED COURSEWORK

Computer and System Sciences: Topics in High Dimensional Geometry; Polynomials and Applications; Topics in the Study of Markov Chains (half-course).

Mathematics: Linear Algebra; Algebra; Probability; Numerical Analysis; Real and Complex Analysis; Measure and Integration; Functional Analysis; Topology; Ordinary and Partial Differential Equations.

PRESENTATIONS AND ACADEMIC ACTIVITIES

- Research presentation: Approximation of Mixed Volumes of Convex Bodies, STCS Symposium, TIFR, August 2025.
- Seminar organization: Catalytic Graph Algorithms, STCS student seminar, TIFR, March 2026.
- Convex and Discrete Geometry Summer School, Alfred Renyi Institute, Budapest, August-September 2023; School on Cluster Algebras, ICTS-TIFR, December 2018; Introduction to SageMath, ICT Mumbai and MTTS Trust, May-June 2020.

SCHOLARSHIPS, EXAMINATIONS AND LANGUAGES

- TIFR / Department of Atomic Energy research scholarship (2019-2026); DST INSPIRE scholarship (2015-2019).
- Qualified JAM in Mathematics and Mathematical Statistics (2018, 2019); ISI M.Math admission test (2018); CMI M.Sc. Data Science written test (2019).
- Languages: Bengali, English, Hindi (native).